Supply Function Equilibrium Bidding Strategies With Fixed Forward Contracts

Hui Niu, *Member, IEEE*, Ross Baldick, *Senior Member, IEEE*, and Guidong Zhu

***Abstract*—During the last 15 years, the regulatory framework for the wholesale sector of the electricity industry has been replaced in many countries by market competition. In such an environment, electric firms compete through both spot market bidding and bilateral contract trading. Firms have to consider their forward contract positions when they make spot market decisions. In this paper, we propose a linear asymmetric supply function equilibrium (SFE) model with transmission constraints to develop firms' optimal bidding strategies considering forward contracts. The characteristics of firms' behaviors are analyzed under the induced equilibrium conditions. The model is applied to the real-time energy market of the Electric Reliability Council of Texas (ERCOT). The effects of forward contracts on the ERCOT market are evaluated. It is shown that the model is able to capture features of bidding behaviors in the ERCOT market.**

***Index Terms*—Bidding strategies, forward contracts, real-time markets, supply function equilibrium (SFE), transmission congestion rights (TCRs), transmission constraints.**

NOMENCLATURE

A. Indices

i	Nodes of the electric network.
j	Firms in the market or nodes of the electric network.
k	Transmission lines.

B. Functions

$C_j(q_j)$	Cost function for firm j.
D_t	System demand curve, including the effect of price.
$N(t)$	Load-time function, representing the load at time $t \in T$ if price were zero.
$q_j(p_t)$	Bid curve for firm j.
$q_{-j}(p_t)$	Collection of supply functions for competitors of firm j.
$T_j(\boldsymbol{p}_t)$	Profit from firm j's TCR.
π_{jt}	Profit function for firm j.

C. Constants

b_{ij}	Absolute value of the branch susceptance between node i and node j.
K	Number of transmission lines in the system.
m	Number of firms.
$P_{\min}$	Minimum bid price cap.
$P_{\max}$	Maximum bid price cap.
T	Each firm submits a bid curve that is assumed to apply throughout time period T.

D. Variables

F_{jT}	Forward contract obligation for firm j during time period T.
P^c_{jT}	Forward contract price for firm j during time period T.
U_j	Generation capacity for firm j.
p_t	Market price at time t.
p_{it}	Market price of node i at time t.
α_j	Intercept of the supply function for firm j.
β_j	Slope of the supply function for firm j.
a_j	Intercept of the marginal cost function for firm j.
d_j	Slope of the marginal cost function for firm j.
γ	Slope of the demand to price relationship.
κ_j	Slope of the inverse supply function for firm j.
r_{ji}	Quantity of firm j's TCR involving node i.

E. Vectors

$\mathbf{B}$	Imaginary part of the nodal admittance matrix.
$\mathbf{D}$	Nodal load vector.
$\boldsymbol{\theta}$	Nodal angle vector.
$\mathbf{p}_t$	Nodal price vector.
$\mathbf{q}$	Nodal generation vector.
$\hat{\mathbf{q}}$	Net nodal injection vector.
$\mathbf{S}$	Shift factor matrix.
$\mathbf{S}_b$	Shift factors matrix for binding transmission constraints.
$\mathbf{FL}_{\max}$	Transmission capacities vector.
$\mathbf{FL}_{b\max}$	Transmission capacities vector for binding transmission constraints.
$\mathbf{z}^*$	Optimal solution to the welfare maximization problem.
$\mathbf{1}$	Vector where all elements are 1.
$\mathbf{1}_j$	Vector with only one nonzero element of 1 at jth element.
$\mathbf{I}_j$	Square matrix with 1 for its jth diagonal element and 0 for other elements.

I. INTRODUCTION

DURING the last 15 years, the previous regulatory framework for the wholesale sector of the electricity industry has been replaced in many countries by market competition. However, a perfectly competitive market does not exist in practice. Many issues can contribute to the market inefficiency, such as market design flaws, market power, and inherent engineering features of power system operations.

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The authors are with the University of Texas at Austin, Austin, TX 78712 USA (e-mail: hniu@ece.utexas.edu; baldick@ece.utexas.edu; gzhu@mail.utexas.edu).

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Some characteristics of electricity markets facilitate the exercise of market power. These characteristics include inelastic demand, limited transmission capacity, and the requirement that supply and demand must balance continuously. Therefore, market power analysis has received attention in both theory and practice. It is important for market designers to fix flaws in market structure and for firms to be able to evaluate their positions and conduct their business.

Noncollusive game theoretical approaches, such as the Bertrand, Cournot, and supply function equilibrium (SFE) models, are widely used to analyze market power issues in electricity markets. Among them, only SFE models [1] reflect the bidding rules in electricity markets, where every firm offers a price and quantity schedule to the market operator. The decision variables in SFE models are the parameters of supply functions, rather than simple price or quantity as in Bertrand and Cournot models. Furthermore, SFE models can also treat cases with zero demand elasticity.

In contrast, the Cournot model requires the residual demand to be elastic. However, the short-run demand elasticity in electricity markets is almost zero. As a result, price predictions from Cournot models depend on assumptions about a competitive fringe and are not reliable. Although the Bertrand model does not require elastic demand, the Bertrand equilibrium is the same as the perfect competitive optimum as long as the market has at least two players with unlimited capacity. This does not match the actual market results.

Neither demand elasticity nor a competitive fringe is necessary for the existence of SFE. The application of the SFE model to the England and Wales (E&W) electricity market [2], [3] shows that SFE models can predict prices that match the empirical data reasonably well.

The above merits have prompted the application of SFE models to various aspects of electricity markets analysis, such as capacity constraints [2], market divestures [2], [3], the relationship between the spot market and contract market [4], [5], strategic behaviors [6], [7], and transmission constraints in a small network with single time period [8], [9]. However, the nonconvexity of the optimization problem faced by each firm means that multiple SFEs or no SFE may exist unless we make restrictive assumptions on the number of firms, the nature of production costs, or supply functions. The costs were restricted to be symmetric in [3] and [10], while constant marginal cost was assumed for generators in [10]. However, for actual markets, the assumption of symmetric costs is questionable. In reality, a generator or a portfolio of generators normally does not have constant marginal cost.

To cope with these problems, asymmetric costs [2], [5], [11] and affine marginal cost functions with equal intercepts [3], [4], [6] or unequal intercepts [2] have been proposed. The advantage of the affine marginal cost function over the more general ones in SFE models lies in the ability to handle asymmetric costs with more than two players. However, restriction to affine marginal cost functions does not guarantee the existence and uniqueness of equilibrium without further limitations on supply functions.

Multiple equilibria become less problematic when practical issues, such as capacity constraints [2], [3], [12], price caps [2], [12], or stability [13] are considered. Capacity constraints tend to limit the range of equilibria. Stability of equilibrium further alleviates this issue since only a stable equilibrium could be sustained in actual markets. In some circumstances, SFEs that are less competitive than the linear SFE are unstable [13].

SFE models have been used to analyze firms' bidding strategies in spot markets. Since SFE models fit actual bidding rules in electricity spot markets better than Cournot models [14]–[17], the predicted behaviors should be more instructive. Hortacsu and Puller [18] and Sioshansi and Oren [19] used SFE models to examine the bidding behaviors in the ERCOT real-time market. Anderson and Xu [20] discuss symmetric duopoly SFE in the case with contracts and price caps. The approaches to mitigate the market power related to transmission congestion rights (TCRs) are discussed with Cournot models in [21] and [22].

In this paper, a linear asymmetric SFE model with transmission constraints is proposed to analyze the bidding strategies with forward contracts. The paper contributes to the literature in several aspects. First, we combine forward contracts, transmission constraints, and multiperiod strategy (an obligation for firms to bid consistently over an extended time horizon such as a day or an hour) into the linear asymmetric SFE framework. As an ex-ante model, it can provide qualitative insights into firms' behaviors. Second, the bidding strategies related to TCRs are discussed by interpreting TCRs as a linear combination of forward contracts in the SFE framework. Third, the model is a general one in the sense that there is no limitation on the number of firms and scale of the transmission network, which can have asymmetric affine marginal costs. In addition to theoretical analysis, we apply our model to the ERCOT real-time market.

The remainder of this paper is arranged as follows. Section II describes the proposed SFE model with fixed forward contracts. Optimal strategies are discussed for both transmission congested and uncongested situations. We briefly describe the ERCOT market structure in Section III. The application of our model to the ERCOT real-time balancing energy market is presented in Section IV. Section V concludes the paper.

II. SFE Model Formulation

Market players have both incentive and ability to affect prices in spot markets. At close to real time, both supply and demand become inelastic. Electricity cannot be stored and generators have limits on how quickly they can be started and ramped up or down. Because the time-varying wholesale price is neither seen nor paid by end consumers, suppliers face a nearly vertical demand curve. Therefore, the profit from market manipulation is greatly increased as the market moves closer to real time. Forward contracts enable buyers and sellers to lock in their prices and quantities in advance of real time, which reduces the profit from increasing real-time prices. Hence, forward contracts play an important role in reducing incentives to exercise market power and hedging the risk related to the high volatility of prices in real-time markets [14], [23]. However, unlike [14], which considers the joint equilibrium in forward and spot prices, we consider only the spot market given particular levels of forward contracts. Based on [2], [4], [5], and [24], the linear SFE model with fixed forward contracts is constructed as follows for

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the transmission uncongested case in Section II-A and for the transmission congested case in Section II-B.

A. Transmission Uncongested Model

Assume there are m firms. The forward contract obligation for firm j during time period T is denoted as F_{jT} at forward price P^c_{jT}. Each firm $j \in \{1, \ldots, m\}$ submits a nondecreasing bid curve that is assumed to apply throughout time period T. The bid curve $q_j : [P_{\min}, P_{\max}] \to [0, U_j]$ is defined by

$$q_j(p_t) = \beta_j(p_t - \alpha_j), \quad \forall\, j. \tag{1}$$

The total production cost function C_j is assumed to be a convex quadratic function

$$C_j(q_j) = 0.5 d_j q_j^2 + a_j q_j, \quad \forall\, j. \tag{2}$$

The system demand curve is assumed to be

$$D_t = N(t) - \gamma p_t. \tag{3}$$

The market clearing condition is

$$\sum_{j=1}^{m} q_j(p_t) = D_t. \tag{4}$$

Since the supply functions are nondecreasing and the market clearing price is the same for all players when there is no transmission congestion, this market clearing condition maximizes the (revealed) social welfare.

The profit for firm j from the spot market is

$$\pi_{jt} = p_t q_j(p_t) - C_j(q_j(p_t)) - F_{jT}(p_t - P^c_T). \tag{5}$$

At equilibrium, each player's optimal supply function is the best response to the supply functions of its competitors over the feasible function space. Neglecting the nondecreasing requirements, the equilibrium q_j^* satisfies

$$\pi_{jt}(q_j^*(p_t), q_{-j}^*(p_t)) \geq \pi_{jt}(q_j(p_t), q_{-j}^*(p_t)).$$

Since forward contracts are decided before the spot market, spot market prices cannot (directly) affect the forward contract obligation F_{jT} and the forward price P^c_{jT}. Differentiating the profit function with respect to price

$$\frac{d\pi_{jt}}{dp_t} = q_j(p_t) + [p_t - C_j'(q_j(p_t))]\frac{dq_j(p_t)}{dp_t} - F_{jT}. \tag{6}$$

Setting (6) equal to zero to obtain the optimal price and using the market clearing condition (4), we have

$$q_j^*(p_t) - F_{jT} = [p_t - C_j'(q_j^*(p_t))]\left(-\frac{dD_t}{dp_t} + \sum_{k\neq j}^{m} \frac{dq_k^*(p_t)}{dp_t}\right).$$

Substituting (1) and (2) into the above equation, we have

$$\beta_j(p_t - \alpha_j) - F_{jT} = (p_t - d_j\beta_j(p_t - \alpha_j) - a_j) \times \left(\gamma + \sum_{k\neq j}^{m} \beta_k\right). \tag{8}$$

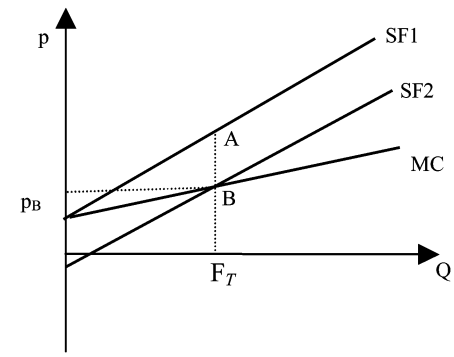

Fig. 1. Supply functions with forward contracts.

A solution to coupled conditions (8) for all firms that satisfies $\beta_j \geq 0$ for all firms is an SFE. Since the bid function is required to be consistent during period T, (8) should be satisfied for the realized prices during that time period. If there are at least two levels of demand cleared during the period T, then there will be at least two prices realized and (8) will be satisfied as an identity [2]. Consequently, the coefficient of price p_t on the left hand and right hand of (8) should be equal, and similarly the constant term should be equal. Therefore, the conditions for an SFE are

$$\beta_j = (1 - d_j\beta_j)\left(\gamma + \sum_{k\neq j}^{m} \beta_k\right), \quad \forall\, j, \tag{9}$$

$$\beta_j\alpha_j + F_{jT} = (a_j - d_j\beta_j\alpha_j)\left(\gamma + \sum_{k\neq j}^{m} \beta_k\right), \quad \forall\, j. \tag{10}$$

Substituting (9) into (10), and rearranging, we obtain

$$\alpha_j = a_j - \left(\frac{1}{\beta_j} - d_j\right) F_{jT}, \quad \forall\, j. \tag{11}$$

From (9) and (11), we observe that the slope β_j of the SFE does not depend on the amount of forward contracts, but the intercept α_j does depend on F_{jT}. When $F_{jT} = 0$, the intercept of a firm's SFE equals the intercept of its marginal cost function

$$\alpha_j = a_j, \quad \forall\, j. \tag{12}$$

Rearranging (11) yields

$$\alpha_j + \frac{1}{\beta_j} F_{jT} = a_j + d_j F_{jT} = C'(F_{jT}), \quad \forall\, j. \tag{13}$$

This means that a firm's SFE will intersect its marginal cost function at the position of its contract obligation, which is consistent with the observations in [4], [5], and [20].

Fig. 1 shows the SFEs for the forward noncontracted (SF1) and contracted (SF2) cases. The marginal cost function is shown in the figure as MC. SF1 is for the noncontracted case, which has the same intercept as the marginal cost function but a different slope. For the forward contracted case, we denote the quantity of forward contracts as F_T, while B represents the point on MC corresponding to F_T. Then SF2 passes through B and

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has the same slope as SF1. In Fig. 1, forward contracts move the SFE prices down by the distance between A and B, which indicates the market power mitigation effects of forward contracts. From Fig. 1, we can see that the greater the amounts of forward contracts, the more mitigation effects. However, we observe that strategic firms bid at a higher price than their marginal cost to sell energy in excess of their contracted quantity and at a lower price than their marginal cost to buy energy from the spot market, which is again consistent with [4], [5], and [20].

B. Transmission Congested Model

1) DC Power Flow Model: Assume there are n nodes and K transmission lines in the system. Firms are indexed by the node numbers of their location. If there are multiple generation firms at a node, artificial nodes could be defined. Therefore, we can assume there is only one firm at each node. Load nodes could be treated similarly to locate them at different nodes from generators. Then, the dc power flow equations, including all buses except the slack bus, are [25]

$$\mathbf{B}\boldsymbol{\theta} = \mathbf{q} - \mathbf{D}. \tag{14}$$

The nodal power balance constraints are implied in this dc power flow model. The real power flow on the branch between node i and j is

$$F_{ij} = b_{ij}(\theta_i - \theta_j). \tag{15}$$

Then transmission capacity constraints are

$$\mathbf{H}\boldsymbol{\theta} \leq \mathbf{FL}_{\max} \tag{16}$$

where $\mathbf{H}$ is the product of the branch susceptance diagonal matrix and an appropriate incidence matrix of branches with nodes. If the kth transmission line connects node i and node j, then

$$h_{ki} = b_{ij}, \quad h_{kj} = -b_{ij}, \quad h_{kl} = 0, \quad \forall\, l \neq i,\ l \neq j.$$

Substituting (14) into (16), we have

$$\mathbf{HB}^{-1}(\mathbf{q} - \mathbf{D}) \leq \mathbf{FL}_{\max}. \tag{17}$$

Define $\mathbf{S} = \mathbf{HB}^{-1}$, whose elements indicate the sensitivity of branch flows to nodal net injections (shift factors). Then the dc power flow and transmission capacity constraints are

$$\mathbf{S}(\mathbf{q} - \mathbf{D}) \leq \mathbf{FL}_{\max}. \tag{18}$$

2) SFE Model With Forward Contracts: We maintain the same assumptions for firms as in Section II-A, except that we use nodal prices, since the nodal price will be different if there is any transmission line congested. In order to facilitate the discussion, we use the inverse form of supply functions

$$p_{jt} = \frac{1}{\beta_j} q_{jt} + \alpha_j = \kappa_j q_{jt} + \alpha_j, \quad \forall\, j \tag{19}$$

where $\kappa_j = 1/\beta_j$ is the inverse of the slope of the supply function.

The nodal demand is

$$D_i = N_i(t) - \gamma_i p_i. \tag{20}$$

Each firm's decision process has to consider the market clearing process of the system operator, which is to maximize system welfare (or minimize negative welfare) constrained by system supply and demand balance (22), transmission capacities (23), and firms' generation limits (24)

$$Min \quad \sum_{i=1}^{n} \left(\int_0^{q_i} p_i(q)dq - \int_0^{D_i} p_i(D)dD \right) \tag{21}$$

$$s.t. \quad \sum_{i=1}^{n} (q_i - D_i) = 0 \tag{22}$$

$$\mathbf{S}(\mathbf{q} - \mathbf{D}) \leq \mathbf{FL}_{\max} \tag{23}$$

$$\mathbf{0} \leq \mathbf{q} \leq \mathbf{q}_{\max}. \tag{24}$$

We assume that when bidding into the spot market, it is known whether or not a line is going to be binding based on the forward contract decisions and other information that becomes available before the spot market. In contrast, if the binding/nonbinding determination is endogenous then, generally speaking, there will not be a pure strategy equilibrium in either the Cournot or SFE framework. Mixed strategy equilibrium approaches could be applied to deal with the endogenous case, but this is beyond the scope of this paper. Here we assume that the information about binding constraints is exogenous. Therefore, we can include only the binding constraints in the formulation of the Lagrangian for the market clearing problem and treat binding constraints as equality constraints. In order to highlight forward contracts and transmission constraints, we do not consider generator capacity limits here. Then the Lagrangian for the above problem is

$$L = \sum_{i=1}^{n} \left(\int_0^{q_i} p_i(q)dq - \int_0^{D_i} p_i(D)dD \right) + \lambda \sum_{i=1}^{n} (q_i - D_i) + \boldsymbol{\mu}_b^T (\mathbf{S}_b \hat{\mathbf{q}} - \mathbf{FL}_{b\max}) \tag{25}$$

where λ and $\boldsymbol{\mu}_b$ are Lagrange multiplier and Lagrange multiplier vector for the system balance constraint and binding transmission constraints. Vector $\hat{\mathbf{q}} = \mathbf{q} - \mathbf{D}$ represents the net nodal injection. By the first-order condition of the Lagrangian (25), the optimal solution $\mathbf{z}^* = [\hat{\mathbf{q}}^{*T}, \lambda^*, \boldsymbol{\mu}_b^{*T}]^T$ satisfies

$$\begin{bmatrix} \boldsymbol{\kappa} & 1 & \mathbf{S}_b^T \\ 1 & \mathbf{0} & \mathbf{0} \\ \mathbf{S}_b & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{q}} \\ \lambda \\ \boldsymbol{\mu}_b \end{bmatrix} = \begin{bmatrix} -\boldsymbol{\alpha} \\ \mathbf{0} \\ \mathbf{FL}_{b\max} \end{bmatrix} \tag{26}$$

where $\boldsymbol{\kappa}$ is a diagonal matrix consisting of the slopes of the inverse SFEs or demand slopes, and $\boldsymbol{\alpha}$ is a vector containing the intercepts of the SFEs or demand function. Equation (26) can be simplified as

$$\mathbf{wz} = \mathbf{g}. \tag{27}$$

According to [24], the sensitivity of the optimal solution $\mathbf{z}^*$ with respect to the parameters of supply function for firm j is

$$\frac{\partial \mathbf{z}^*}{\partial \kappa_j} = -\mathbf{w}^{-1}\mathbf{I}_j\mathbf{z}^* \tag{28}$$

$$\frac{\partial \mathbf{z}^*}{\partial \alpha_j} = -\mathbf{w}^{-1}\mathbf{1}_j. \tag{29}$$

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We have that $\partial q_{jt}^*/\partial\kappa_j = -\xi_{jj}q_{jt}^*$ and $\partial q_{jt}^*/\partial\alpha_j = -\xi_{jj}$, where ξ_{jj} is the jth diagonal element of $\mathbf{w}^{-1}$.

The profit function for firm j is

$$\pi_{jt} = (\kappa_j q_{jt} + \alpha_j)(q_{jt} - F_{jT}) - 0.5 d_j q_{jt}^2 - a_j q_{jt} + F_{jT}P_T^c. \quad (30)$$

The first-order conditions for firm j's profit are

$$\frac{\partial \pi_{jt}}{\partial \kappa_j} = q_{jt}^{*2} - \left[(2\kappa_j - d_j)q_{jt}^* + (\alpha_j - a_j - F_{jT}\kappa_j)\right] \times \xi_{jj}q_{jt}^* - F_{jT}q_{jt}^* = 0 \quad (31)$$

$$\frac{\partial \pi_{jt}}{\partial \alpha_j} = q_{jt}^* - \left[(2\kappa_j - d_j)q_{jt}^* + (\alpha_j - a_j - F_{jT}\kappa_j)\right] \times \xi_{jj} - F_{jT} = 0. \quad (32)$$

Comparing (31) to (32), we observe that (31) will be satisfied as long as (32) is satisfied. Similar to the uncongested case, we assume that at least two different dispatches are realized during period T. Then by setting, respectively, the coefficients of q_{jt}^* and the constant term of (32) to zero and rearranging, we obtain equilibrium conditions

$$1 - (2\kappa_j - d_j)\xi_{jj} = 0, \quad \forall j \quad (33)$$

$$\alpha_j = a_j - F_{jT}(\kappa_j - d_j), \quad \forall j \quad (34)$$

where we have used (33) to simplify the terms in (34). By comparing (33) and (9), we observe that the transmission *constraints* affect the slopes of the SFEs. However, forward *contracts* do not affect the slopes of the SFEs for both the transmission uncongested and the transmission congested cases. Comparing (11) and (34), we observe that the intercepts are similar, which means forward contracts have similar effects on the intercepts of SFEs for the transmission congested case as for the transmission uncongested case that was discussed in Section II-A. Therefore, we will not discuss the intercepts again here.

An iterative method can be developed based on (33) and (34) to calculate the SFEs with transmission congestion under forward contracts. Nevertheless, we should note that the binding transmission constraints are assumed not to change during the iterations. Otherwise, this method will fail to converge, indicating that the pure SFE does not exist.

3) Transmission Congestion Rights: In electricity markets, transmission congestion rights (TCRs) are a special form of contract provided for market participants to hedge the risk of congestion prices. TCRs pay based on the difference between nodal prices or the shadow price on the congested line. The profit from firm j's TCRs can be expressed as

$$T_j(\mathbf{p}_t) = \sum_{i=1}^{n} r_{ji}p_{it}. \quad (35)$$

If the net TCR holdings are from other nodes to node i, then r_{ji} is positive. Otherwise, it is negative.

The profit function for firm j with TCRs is

$$\pi_{jt} = (\kappa_j q_{jt} + \alpha_j)q_{jt} - 0.5 d_j q_{jt}^2 - a_j q_{jt} + T_j(\mathbf{p}_t). \quad (36)$$

The conditions for SFE can be obtained following similar steps to the case for forward contracts. That is, analogously to (31) and (32), we have

$$[1 - (2\kappa_j - d_j)\xi_{jj}]q_{jt}^{*2} + r_{jj}q_{jt}^* - (\alpha_j - a_j + r_{jj}\kappa_j)\xi_{jj}q_{jt}^* - \sum_{i\neq j}^{n} r_{ji}\kappa_i\xi_{ij}q_{it}^* = 0 \quad (37)$$

$$[1 - (2\kappa_j - d_j)\xi_{jj}]q_{jt}^* + r_{jj} - (\alpha_j - a_j + r_{jj}\kappa_j)\xi_{jj} - \sum_{i\neq j}^{n} r_{ji}\kappa_i\xi_{ij} = 0 \quad (38)$$

where ξ_{ij} is the element of $\mathbf{w}^{-1}$ at (i, j) position. By (37) and (38), we found that the slope and intercept of a firm's SFE are affected by its TCR shares. Here we only discuss the simplest case to illustrate possible strategies related to TCRs. It is assumed that $\xi_{ij} = 0, \forall\, i \neq j$. This assumption means firm j has very little effect on the prices of nodes except the node where it is located. Therefore, we can ignore the terms related to nodes other than node j in (37) and (38), and the slope of SFE is the same as (33), but the intercept changes to

$$\alpha_j = a_j + r_{jj}(\kappa_j - d_j), \quad \forall j. \quad (39)$$

As a result, if $r_{jj} > 0$, it may be profitable for the firm to increase the intercept of its SFE to increase the TCR value by increasing the shadow price. In contrast, if $r_{jj} < 0$, it may be profitable for the firm to decrease its intercept.

Now we assume that firm j owns generation units on both node j and node k and that firm j possesses a TCR from node j to node k. We now assume that the firm cannot significantly affect the prices at nodes $i \neq j, k$. The shadow price on line from node j to node k is related to $(p_k - p_j)$ so that $r_{jj} < 0, r_{jk} > 0$. Analogously to the previous case, the firm may increase the intercept of its SFE at node k and decrease its intercept of SFE at node j to increase the shadow price on the line and its profit.

III. ERCOT Electricity Market

The ERCOT market began to operate on July 31, 2001, under the ERCOT independent system operator (ISO), which oversees the operation of approximately 77 000 megawatts of generation. Unlike most of the other restructured markets in the United States, which have both a day-ahead energy pool market and a real-time market, currently ERCOT has only a real-time balancing energy service (BES) market. However, ERCOT has a day-ahead portfolio energy schedule process and day-ahead ancillary services markets. We only focus on the BES market in this paper. Other details about the market are available in [26].

A. Schedule Process

The schedules in ERCOT indicate the matching of generation resources with load serving entities (LSEs). LSEs forecast their customer load and negotiate privately with generation resources (PGCs) to buy energy in order to supply their customers. Forward contracts between them are typically incorporated into their schedules, which represent about 95%–97% of delivered

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energy in ERCOT. Market participants are required to submit their quarter-hourly schedules to the ERCOT ISO through qualified scheduling entities (QSEs).

B. Real-Time Balancing Energy Markets

The ERCOT ISO dispatches deviations of generation and load from the scheduled generation and load. For every 15-min interval, the BES market is cleared based on the hourly balancing energy up (BEU) and balancing energy down (BED) bids and schedules submitted by QSEs. BEU and BED bids are specified by congestion zone with monotonically increasing ordered pairs of prices and cumulative megawatts ($/MWh, MW). The bid caps are $1000/MWh for BEU and −$1000/MWh for BED. Besides the zonal portfolio level BES bids, QSEs could also voluntarily submit their resource specific premium bids to help solving local reliability issues.

C. Congestion Management

The ERCOT ISO categorizes congestion as either interzonal congestion or local congestion. The transmission grid, including generation resources and loads, is divided into several congestion zones determined annually. Each congestion zone is defined such that each generation resource or load within the zone has a similar effect (characterized by its "shift factor") on the transmission facilities between congestion zones, which are called commercially significant constraints (CSCs). The congestion on CSCs or predefined closely related elements (CREs) is called interzonal congestion or CSC congestion. Congestion on other transmission elements is viewed as local congestion.

Prior to June 2004, ERCOT used a two-step congestion management scheme. CSC congestion is managed first. The market clearing price of energy (MCPE) is determined for each zone based on zonal BES bids, CSC transmission limits, and zonal generation-weighted average shift factors. Therefore, generators are exposed to locational prices that reflect the average effect of location in zones on CSCs. TCRs and preassigned congestion rights (PCRs) were implemented as financial hedges against the CSC congestion cost on February 15, 2002. TCRs are awarded in yearly and monthly TCR auctions. If there is local congestion, the ERCOT ISO relies on unit-specific bids and shift factors to instruct particular resources to relieve local congestion. The unit specific dispatches are settled based on unit specific bids or at the "out of merit" price set by protocols, rather than by MCPEs. The cost of relieving local congestion is uplifted to each QSE based on its load ratio share (LRS).

IV. Application

Since most of the energy is traded through bilateral contract markets in the ERCOT market, it is instructive to apply the proposed SFE model to the ERCOT real-time BES market to gain insight about bidding behaviors of market participants with forward contracts. We apply our model to the ERCOT for the time from January 2002 to April 2003.

A. Assumptions

1) Forward Contract: Since there is no public resource about firms' actual forward contract positions in ERCOT, we assume schedules submitted by QSEs match forward contracts between generation and LSEs they represent. However, schedules could differ from forward contracts. In this case, our SFE model may underestimate or overestimate market power.

TABLE I
BID AND DISPATCH SHARES

QSE Name	BEU Shares		BED Shares	
	Bid	Dispatch	Bid	Dispatch
TXU	44.7%	24.8%	25.1%	13.1%
Texas Genco	24.8%	21.2%	28.2%	35.1%
Calpine	5.1%	14.6%	7.3%	12.4%
AEP	8.6%	8.8%	9.8%	3.3%
Others	16.9%	30.8%	29.6%	36.2%

2) Strategic Players: Since the schedules and bids in ERCOT are submitted by QSEs, we identify QSEs as the market players even though a QSE can represent more than one firm. Actual BES bid and dispatch shares are used to identify major players. The bid (dispatch) share is the percentage of the bid (dispatched) quantity from a specific QSE divided by the total system bid (dispatched) quantity, which is based on hourly bid curves submitted and actual first-step dispatch, by which MCPEs are decided. Table I shows the average bid and dispatch shares for the four biggest QSEs during the test period.

These four major QSEs represent the five biggest firms in ERCOT, including TXU, Reliant Energy, Calpine, City Public Service of San Antonio (CPS), and Central Power and Light (CPL). Texas GENCO represents both reliant and CPS. American Electric Power (AEP) includes CPL in its portfolio. These five firms account for about 24%, 18%, 8%, 6%, and 5% of the total system installed capacity, respectively. These firms also account for most of the installed capacity of the QSEs representing them. All other QSEs are treated as one strategic player in our SFE model.

3) Demand: We assume there is no demand elasticity in the real-time market and use the actual BES demand for each interval during our test period for the demand in the model.

4) Marginal Cost Functions: The production cost of each major QSE is based on the units within its portfolio. Since more than 90% of QSEs' installed capacities consist of fossil fuel units, we only consider fossil fuel units to construct their production cost function. The output from other kinds of units, including nuclear, hydro, and wind units, are deducted from the respective portfolio schedules, and their effects on CSCs are considered based on their output.

For fossil fuel generation units, the production cost for each unit is based on the fuel price and its average heat rate, which is a common assumption in the literature. Although incremental unit heat rate would be more appropriate, unfortunately this information is not available to us. The average heat rate for each unit in ERCOT is obtained from [27]–[29]. The gas prices are obtained from the market oversight division (MOD) at the Public Utility Commission of Texas (PUCT). Coal prices are from [30]. The capacity of a unit is based on its installed capacity multiplied by one minus its forced outage probability [31].

The intercept and slope for a QSE's marginal cost function are obtained by a linear approximation of its marginal cost curve based on the fuel price and heat rate information. For the QSEs

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with both coal and gas units, marginal cost functions are constructed for coal and gas units, respectively. No attempt is made to capture the operating, planned maintenance, environmental cost, and ancillary services purchased in the ERCOT day-ahead market. Those factors could make us underestimate the marginal cost.

5) Capacity Limits: In the ERCOT BES market, the maximum BEU and BED capacity a player can supply is its installed capacity minus scheduled generation and its scheduled generation, respectively. It is found that QSEs bid only part of their BES capacity, about 15% for BEU and 30% for BED on average during the test period. This is not necessarily physical withholding, which is one expression of market power. The data shows that only about 17% and 23% on average of the BEU and BED bid quantity were actually dispatched during our test period. Firms might have other reasons for not bidding all their BES capacity, such as unit commitment decisions, operation preference, and fuel supply. Therefore, to avoid detailed considerations of these factors, we assume that the actual bid quantities of each QSE are their actual available quantity. If we relax bid limits based on installed capacity and schedules, the SFE prices change by less than 3%, because normally only part of the BES bid quantity was actually dispatched.

6) Transmission Congestion: Transmission congestion is a source of market power. In the ERCOT SFE model, we consider only the CSC congestion, because MCPEs depend on CSC congestion, not local congestion during our test time period. Monthly zonal weighted average shift factors are used to represent the effect of zonal generation and load on CSCs. The ERCOT ISO publishes the possible binding constraints before real-time based on the information submitted to the ISO from QSEs. We assume that the real-time market has a small enough effect so that no expected nonbinding constraints will become binding due to balancing dispatch. Therefore, we assume that it is known in advance whether or not a line is going to be constrained in the real-time market and only include binding constraints in (26) for the ERCOT application.

Although allocations of TCRs and PCRs can affect market power related to congestion as we discussed in Section II, there is no public resource about them in ERCOT. We assume a QSE's TCR shares are consistent with its forward contracts. However, its actual TCR or PCR shares may be higher or lower than its forward contracts, which may make us underestimate or overestimate market power for the congested cases. The generators of each major player in each zone are treated as one player for the congested case.

B. Results

In the application, a supply function is formed for each of the major players hourly based on (9) and (11) for transmission uncongested hours and (33) and (34) for transmission congested hours during January 2002 to April 2003. For every 15-min interval, the market is cleared according to (4) or (21)–(24). When some BEU bid prices are lower than some BED bid prices (called overlap) [32], they may cancel against each other to ensure that most efficient sources are dispatched. There was overlap for about 43% of intervals during the test period.

TABLE II
RESULT OF SFE TRANSMISSION UNCONGESTED CASE

Month	BEU ($/MWh)				BED ($/MWh)		
	MC	SFE_NC	SFE_C	Actual	MC	SFE_C	Actual
Jan-02	21.2	24.9	21.0	21.1	22.0	20.8	7.7
Feb-02	21.7	27.9	21.7	21.1	22.1	20.8	9.9
Mar-02	28.2	34.6	28.2	32.8	27.5	26.4	13.2
Apr-02	33.7	40.5	33.9	35.9	31.5	29.5	17.5
May-02	32.7	38.8	33.6	34.5	33.0	31.8	15.2
Jun-02	33.2	41.6	34.6	33.7	31.7	30.6	16.8
Jul-02	30.8	39.4	32.4	32.9	29.3	28.3	16.6
Aug-02	30.9	39.9	32.5	30.5	31.2	30.4	17.2
Sep-02	34.9	42.4	35.7	31.8	33.8	33.2	15.0
Oct-02	37.8	43.7	38.3	41.7	34.4	33.8	17.2
Nov-02	36.5	40.5	36.4	35.4	34.1	34.5	14.0
Dec-02	43.6	48.8	43.8	40.2	40.0	40.4	11.7
Jan-03	51.8	60.2	53.0	52.5	48.4	47.8	17.7
Feb-03	86.7	101.5	92.9	103.5	46.4	48.2	19.3
Mar-03	67.1	76.3	69.6	71.8	42.4	41.6	30.7
Apr-03	50.6	58.5	52.5	57.9	40.7	43.5	15.2
WAP	49.1	57.6	51.1	53.2	33.4	32.6	16.2
PC Margin	-	17.3%	4.0%	8.2%	-	2.4%	-51.6%

Some intervals were excluded from the calculation to avoid the influence of abnormal events, such as forced outages, software failures, and operational errors [33]. There are a total of 464 intervals excluded from the total 46 560 intervals for the testing time period, which account for only about 1% of the total test intervals.

The simulation was performed with Matlab 6.5 on a Linux workstation with a 1.8-GHz Xeon processor. All information related to congestion, transmission limits, actual market prices, demand, and schedules are available in [33]. The computation took less than 0.1 s on average for an uncongested case and about 28 s on average for a congested case.

1) Transmission Uncongested Case: There are about 83% uncongested intervals during our test period. The monthly weighted average marginal cost (MC), estimated SFE prices for noncontracted case (SFE_NC) and contracted case (SFE_C), and actual market prices for the uncongested cases are listed in Table II. Table II also shows the weighted average of MC and SFE prices (WAP) and the percentage of price-cost margin (PC Margin) for the uncongested cases over the test period. Since the no-contracted case is a pure pool market case, only BEU would be dispatched in such a market. Therefore, the no-contracted case is only listed for the BEU market.

From Table II, we see that forward contracts decrease SFE price-cost margin by 13.3%, from 17.3% to 4.0%, which indicates the market power mitigation effect of forward contracts. The counterfactual SFE BEU noncontracted prices are usually higher than both the SFE contracted prices and the actual market prices. Comparing contracted SFE prices with actual market prices, we find that average estimated SFE BEU contracted price of $51.1/MWh matches the average actual market price $53.2/MWh well, with a -3.9% difference between them.

Several factors may contribute to the difference between the estimated contracted SFE and actual prices. First, firms' actual bids could involve a much more complicated asset optimization process than the SFE model. A firm can include their fuel supply, environmental constraints, forced or planned outage, volatility of demand, and operational issues into consideration.

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TABLE III
RESULT OF SFE TRANSMISSION CONGESTED CASE

Month	BEU ($/MWh)				BED($/MWh)		
	MC	SFE_NC	SFE_C	Actual	MC	SFE_C	Actual
Jan-02	21.1	24.5	21.0	24.2	22.3	19.7	5.0
Feb-02	21.6	27.1	28.0	35.8	21.8	19.3	9.1
Mar-02	28.3	35.2	28.8	30.5	27.9	24.2	12.3
Apr-02	35.5	45.6	37.2	79.8	32.6	28.3	17.3
May-02	33.6	39.2	34.0	33.3	35.2	31.2	16.7
Jun-02	33.3	42.4	35.6	36.8	32.4	32.0	19.5
Jul-02	30.7	39.6	33.0	33.6	29.6	28.7	16.4
Aug-02	32.6	43.1	35.0	34.3	31.4	31.0	19.6
Sep-02	35.5	45.2	38.3	38.5	35.7	35.0	18.5
Oct-02	40.1	51.4	43.4	52.6	36.3	31.2	23.2
Nov-02	33.0	39.9	39.0	40.1	30.4	19.2	15.8
Dec-02	43.9	49.0	44.5	41.4	41.7	43.8	29.5
Jan-03	54.6	65.2	57.0	52.2	49.7	47.7	20.5
Feb-03	75.1	85.3	79.0	63.9	57.7	56.9	27.0
Mar-03	64.6	73.1	67.2	86.2	47.3	45.2	36.7
Apr-03	50.1	59.0	52.8	74.8	47.6	45.5	22.9
Average	35.6	43.4	37.9	43.1	31.3	29.0	16.1
PC Margin	-	21.8%	6.4%	20.8%	-	7.3%	-48.6%

Second, their actual marginal costs and bid curves could be more complicated than our linear supply functions. Third, we assume each QSE is one player, which may actually include several competing firms. In addition, the ERCOT ISO considers operational constraints such as ramp rate in the market clearing process, which we do not consider because this information is not available publicly. The relatively low price cost margin (8.2%) of actual BEU price suggests an efficient BEU market performance.

In contrast, the average SFE BED price of $32.6/MWh does not fit the actual BED price of $16.2/MWh well. The relatively lower actual BED price indicates that some participants do not want to decrease their generation output from their schedule level in the real-time market, even though they can buy cheap energy there to supply their contract obligation. The relatively low negative price cost margin (-51.6%) of the actual BED price suggests an inefficient BED market performance.

2) Transmission Congested Case: There are about 17% transmission congested intervals during our test period. Results for congested intervals are shown in Table III. Table III shows similar market mitigation effects of forward contracts, decreasing SFE price-cost margin by 15.4% from 21.8% down to 6.4%, as observed for the transmission uncongested cases. The difference between the average estimated contracted SFE price of $37.9/MWh and actual BEU price of $43.1/MWh is about -11.9%, while the difference between the average SFE BED price of $29.0/MWh and the actual BED price of $16.1/MWh is 80.4%. These differences between the estimated contracted SFE prices and the actual market prices are somewhat larger than for the transmission uncongested cases, and the price cost margins are slightly different.

Several factors could contribute to the different margins. CSC congestion, as discussed in Section II, is an obvious one. In addition, we did not consider the effects of actual TCR or PCR assignment because this information is not available publicly. However, as we discussed in Section II, players may increase or decrease their BES bids to maximize their profit from both energy trading and TCRs. Then the SFE prices considering actual TCRs could be higher for the BEU and lower for the BED than the contracted SFE prices shown in Table III. Therefore, if the TCR informations were available, the match between our SFE results and actual market prices for the congested case may be improved.

Though contracted SFE prices deviate from actual market prices by different amount for the transmission congested cases and for the transmission uncongested cases, the overall contracted SFE BEU weighted average price of $49.0/MWh still matches the actual BEU price of $51.6/MWh well for the test period since the congested intervals account only about 17% of the time during the test period. The estimated SFE BED average price of $31.9/MWh is not close to the actual BED price of $16.2/MWh. The overall price cost margin for the actual BEU and BED prices are 9.8% and -50.8%, which suggests relatively efficient BEU and inefficient BED market performance, respectively, in ERCOT during the test time period.

3) Discussion: Several issues can contribute to the relatively efficient BEU market. The high volume of forward contracts decreases the incentive of major market players to raise real-time market prices. If forward contract volume decreases, the real-time BEU price may increase. The high capacity reserve margin of ERCOT (34% for 2002 and 21% for 2003) presumably also contributes to the relatively efficient BEU market performance.

The inefficient performance of BED market may be caused by the following factors. Technical limits, such as unit commitment, low generation limits, and operational constraints, can result in relatively high adjustment costs that make firms not willing to decrease their generation levels. This, on the other hand, may also indicate the inefficiency of their forward decisions or difficulty in adjusting their forward decisions. Currently ERCOT is undergoing a market redesign process. A day-ahead energy market will be implemented for the new market, which will be helpful to improve the efficiency of forward decisions. Furthermore, some firms may not want to expose themselves to the real-time market opened three years ago and hold their capacity to cover their native load in case of any unexpected situations. Another possibility is that the small profit from the low volume real-time market does not outweigh the cost for changing their previous practices before the ERCOT real-time market opened.

V. CONCLUSIONS

An asymmetric linear SFE model with transmission constraints is proposed to develop firms' optimal bidding strategies given their forward contracts. The proposed model is applied to the real-time balancing energy market of ERCOT from January 2002 to April 2003. Market power mitigation effects of forward contracts are evaluated. Contracted SFE prices are observed to match the actual market prices of balancing up energy well. However, the actual average balancing down energy prices are found to be much lower than the contracted SFE prices, which indicates that some market participants did not choose to buy available cheap energy in the real-time market to supply their contract obligations. Adjustment costs, hedging risk costs, and other individual considerations may contribute to the unwillingness of participants to decrease their generation output in

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the real-time market. Competitive forward contract markets and a day-ahead market will be helpful to increase the efficiency of market participants' forward decisions. Our model can be used to analyze the effect of different forward contract level on real-time market prices. It can also be extended to analyze the interaction between day-ahead market and real-time market or the market power related to TCRs in the future.

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Hui Niu (M'03) received the B.E. and M.E. degrees from Harbin Institute of Technology, Harbin, China, and the Ph.D. degree from the Electrical and Computer Engineering Department, the University of Texas at Austin.

Her main research interest is in electricity markets.

Ross Baldick (M'91–SM'94) received the B.Sc. degree in mathematics and physics and the B.E. degree in electrical engineering from the University of Sydney, Sydeney, Australia, and the M.S. and Ph.D. degrees in electrical engineering and computer sciences in 1988 and 1990, respectively, from the University of California, Berkeley.

From 1991 to 1992, he was a Postdoctoral Fellow at the Lawrence Berkeley Laboratory. In 1992 and 1993, he was an Assistant Professor with Worcester Polytechnic Institute, Worcester, MA. He is currently a Professor in the Department of Electrical and Computer Engineering, the University of Texas at Austin.

Guidong Zhu received the B.E. and Ph.D. degrees in engineering mechanics from Harbin Institute of Technology, Harbin, China, and the Ph.D. degree in management science from the University of Texas at Austin.

His main research interest is in business applications of operations research methods.

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